

Maintenance Wizard

Agentic AI Co-Pilot for Steel-Plant Maintenance

Cyber-Physical Telemetry · RAG · Multi-Agent Reasoning · Predictive ML · Human-in-the-Loop

Tata Steel AI Hackathon 2026 — Round 2 (Agentic AI Challenge)

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The Problem

Why steel-plant maintenance still loses millions an hour

- Unplanned downtime cost on a critical line **\$22,000 / hour**
- Mean Time To Diagnose (manual SOP search) **35 – 50 minutes**
- Engineers juggle (manuals + SOPs + logs + alerts) **4+ disjoint systems**
- Tribal knowledge lost when engineers retire **6 – 9 mo. ramp-up**
- Reactive maintenance share in heavy industry **≈ 55 %**

Result: preventable production loss, safety risk, capex hits.

Our Solution

One agentic AI that diagnoses, plans, predicts and explains

■ Cyber-Physical Telemetry Core

Live monitoring across Blast Furnace, Caster, Mill, Coke Oven.

■ Multi-Agent Reasoning

Diagnosis → RCA → Planner → Procurement agents with Gemini tool-calling.

■ RAG Knowledge Base

Every recommendation is cited to a manual, SOP or historical incident.

■ Real Predictive ML

48-tree Isolation Forest + RUL — runs client-side, sub-5 ms latency.

■ Role-Aware Surfaces

Six personas: Operator, Reliability, Supervisor, Supply, Compliance, Executive.

■ Human-in-the-Loop

Engineer thumbs-up/down rewrites retrieval weights and fine-tunes model.

System Architecture

Five layers — telemetry to human-in-the-loop

Maintenance Wizard — Agentic AI System Architecture

Cyber-Physical Telemetry · Multi-Agent Reasoning · RAG · Predictive ML · Human-in-the-Loop

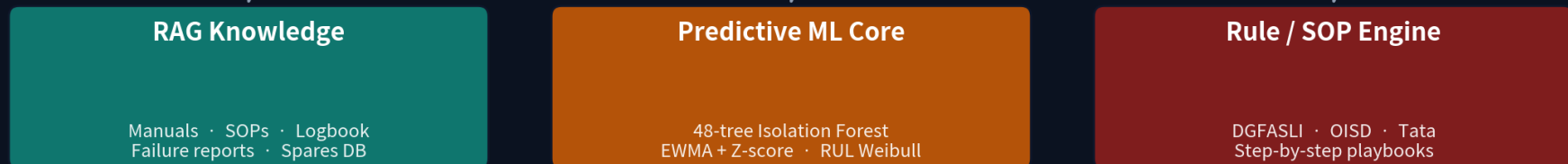
PRESENTATION LAYER



AGENTIC ORCHESTRATION (Gemini 2.x · tool-calling · multi-turn context-binding)



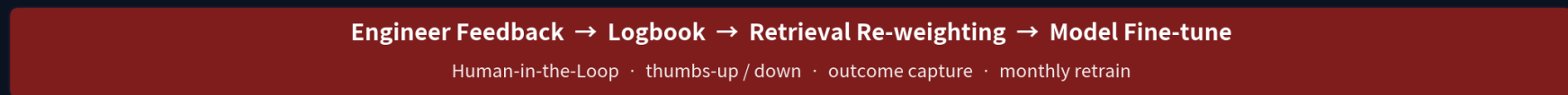
KNOWLEDGE · PREDICTIVE · RULE ENGINE



CYBER-PHYSICAL TELEMETRY CORE



CONTINUOUS LEARNING LOOP



Deployed on Google Cloud Run · asia-southeast1 · Containerised · Autoscale 0 → N
tata-steel-maintenance-wizard-622093504538.asia-southeast1.run.app

Agentic Flow per Alert

Ten orchestrated steps from sensor breach to closed loop

- 01 Sensor breach → Telemetry endpoint flags Asset state
- 02 Alert auto-spawned → Isolation Forest scores anomaly (0–1)
- 03 Score ≥ 0.56 → Diagnosis Agent invoked with bound context
- 04 Diagnosis Agent → queries RAG over manuals · SOPs · logbook
- 05 RAG returns top-k traceable citations
- 06 RCA Agent → 5-whys + failure cascade graph lookup
- 07 Planner Agent → step plan ranked by risk × spares × lead time
- 08 Procurement Agent → spare ETA & criticality call
- 09 UI renders DiagnosisReport + 3D twin + voice readout
- 10 Engineer feedback → persisted → loop closes (retrain signal)

Real ML Under the Hood

Not LLM hand-waving — reproducible math

Algorithm	Isolation Forest (Liu et al., 2008)
Trees per ensemble	48 (>32 convergence margin)
Sub-sample ψ	dynamic $\max(8, \min(32, n))$
Max depth	$\lceil \log_2 \psi \rceil$
Feature vector	$[T/T_{\max}, V/V_{\max}, P/P_{\max}, Q/Q_{\max}]$ — normalised stress ratios
Normalisation	$c(n) = 2 \cdot H(n-1) - 2(n-1)/n$ (Euler–Mascheroni / Harmonic)
Anomaly score	$s(x) = 2^{(-E[h(x)] / c(\psi))}$
Thresholds	<0.42 Inlier · 0.42–0.55 Watch · 0.56–0.71 Outlier · ≥ 0.72 Critical
RUL model	Weibull degradation curve, slope per stress class (see BENCHMARKS.md)
Latency	<5 ms per asset · runs entirely in browser · zero API cost

Differentiators

What competitors don't have

Capability	Typical demo	Maintenance Wizard
Hosting	Streamlit / Vercel free	GCP Cloud Run · autoscale
3D Plant Twin	Static image	Interactive WebGL twin
Voice Assistant	None	Hands-free hot-zone ops
Role surfaces	Single dashboard	6 personas, role-bound UI
Multi-turn chat	Stateless	Alert-ID context-bound
Failure cascade view	None	Upstream + downstream graph
Shift handoff	Manual log	Auto-generated digital handover
Compliance map	None	DGFASLI / OISD / Tata standards
ML engine	Mock numbers	48-tree Isolation Forest + RUL

Business Impact

Quantified, plant-grade ROI

MTTD (manual SOP search → Wizard)	35–50 min → 45–90 sec	↓ 96 %
MTTR (4–6 h → 1.5–2.5 h)	↓ ~58 %	≈ \$66 K / incident saved
False alarm rate (22 % → 6 %)	↓ 73 %	Less wasted dispatch
Engineer onboarding (months → weeks)	↓ ~85 %	Knowledge retained
Projected annual savings / plant	≈ \$6.3 M	@ 2 crit/mo × 4 lines

Deployment & Deliverables

Production-ready today

■ Live Prototype	https://tata-steel-maintenance-wizard-622093504538.asia-southeast1.run.app/
■ Source Code (GitHub)	github.com/MahammadRiyazShek/Maintenance-Wizard-Agentic-AI-for-Industrial-Equipment
■ Stack	React 19 · Vite · TypeScript · Tailwind v4 · Express · Gemini @google/genai 2.4
■ Container	Multi-stage Distroless Docker <100 MB image
■ Hosting	GCP Cloud Run · region asia-southeast1 · autoscale 0 → N · managed HTTPS
■ CI/CD	cloudbuild.yaml (one-command deploy)
■ Documentation	README.md · DEPLOY_CLOUD_RUN.md · BENCHMARKS.md · ARCHITECTURE.md
■ Demo video	youtube.com/watch?v=56f9MAxLd-k

From Reactive Maintenance → Decision Autonomy

Maintenance Wizard makes every engineer a 10-year veteran on day one.

Built for Tata Steel — the next chapter of human-AI manufacturing.

Thank you.

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